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# Naïve Bayes Implementation with Confusion Matrix and Accuracy

import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

# 1. Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# 2. Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 3. Initialize Naïve Bayes classifier
model = GaussianNB()

# 4. Train the model
model.fit(X_train, y_train)

# 5. Predict on test data
y_pred = model.predict(X_test)

# 6. Evaluate performance
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nAccuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred, target_names=iris.target_names))

# 7. Predict a new sample
new_sample = np.array([[5.1, 3.5, 1.4, 0.2]]) # Example flower
prediction = model.predict(new_sample)
print("Predicted Class for new sample:", iris.target_names[prediction][0])
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Confusion Matrix:
[[10  0  0]
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[ 0  9  0]  
[ 0  0 11]]
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Accuracy: 1.0

Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Predicted Class for new sample: setosa